# Systems Analysis and Design (CO3115)

# Software Requirements Specification

**for**

# Online Selling System (OSS)

**Version 1.0 approved**

**Prepared by Le Minh Trung**

**Luu Quang Hoang Cuong**

**Mai Hoang Phuc**

**Ho Chi Minh City University of Technology (HCMUT) – VNU-HCM**

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# Revision History

| Name | Date | Reason For Changes | Version |
|---|---|---|---|
| Initial version | 25/10/2025 | Initial version of the SRS | 1.0 |
| SDS Design | 23/11/2025 | Add the architectural design and database design | 1.0 |

# 1. Introduction

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements for the Online Selling System (OSS) for the Electronics Store, Version 1.0.

This SRS describes the full scope of the system, including modules for product browsing, ordering, purchasing and receiving, along with system maintenance for admin and product management for staff. The document covers both customer and administrative functionalities that are required to digitalize the store's current manual sales process.

This document serves as a guideline for the development team, testers, and stakeholders to ensure a shared understanding of system expectations and constraints.

## 1.2 Document Conventions

This document follows IEEE 830 – Software Requirements Specification Standard:

- Headings and numbering indicate hierarchical sections for easy reference (e.g., 1.1, 1.2, 1.3).
- Bold text is used for names, feature titles, and key terms.
- Italics is used to highlight examples or important notes.
- Functional requirements are written in clear, testable statements using the format "The system shall…" to indicate mandatory behaviors.
- Priorities are explicitly assigned to requirements as High, Medium, or Low, and are not inherited unless stated otherwise.

## 1.3 Intended Audience and Reading Suggestions

This document is intended for other contributors involved in the development and deployment of the Online Selling System:

- Developers: to better understand the system functionalities, design constraints, and implementation requirements.
- Project Managers: to plan resources, timelines and monitor project progress.
- Marketing staff: to align features with go-to-market strategy and customer personas, provide feedback on user journeys to inform feature prioritization and launch planning.
- Testers/QA Team - to design and execute test cases that comply with both functional and non functional requirements.

- Store staff (end users): to understand the system's capabilities and constraints in managing products, promotions and orders.
- Store administrator (end users): to understand the system's capabilities and constraints in staff management, system updates and system settings management.
- Customers and recipients (end users): to gain an overview of the system features relevant to their shopping experience.

Readers are advised to follow the numerical order of the sections for a logical and comprehensive understanding of this document. Each section is summarized as follows:

- Section 1 – Introduction: Provides a general overview of the document and the project, including its purpose, scope and intended audience. It also covers the document conventions and references.
- Section 2 – Overall Description: Describes the users' needs, the system context and overall features, operating environment and relevant constraints that influence system design and implementation.
- Section 3 – Data Flow Diagram: Presents the visual representation of data movement within the system, illustrating major processes, data stores, and data interactions among system components.
- Section 4 – System Features: Details the functional requirements of the system, outlining specific features and behaviors that support the intended operations and user interactions.
- Section 5 – Non-Functional Requirements: Defines non-functional attributes such as performance, security, usability, reliability and maintainability that the system must satisfy.
- Section 6 – Other Requirements: Specifies any additional requirements, including legal, regulatory, or compliance obligations relevant to the system.
- Section 7 – System Analysis: Provides an analytical view of the system's structure and behavior, including use case diagrams, use case scenarios and other related diagrams.
- Section 8 – Conclusion: Summarizes the key points of the document and restates the overall objectives and expected benefits of the system.

## 1.4 Project Scope

The Online Selling System (OSS) is designed to digitalize and modernize the store's sales operations for electronic products, including smartphones, televisions, and computers. The system aims to provide customers with a seamless online

shopping experience, featuring product browsing, filtering, purchasing, flexible payment and delivering options.

Key objectives includes:

- Enhancing customer convenience and satisfaction through online accessibility and comfortable shopping experiences.
- Streamlining store operations by automating product management and order handling.
- Expanding market reach through digital channels.
- Reducing manual workload and operational inefficiencies.

### 1.5 References

*None*

## 2. Overall Description

### 2.1 Product Perspective

The Online Selling System (OSS) is a new, self-contained web-based application designed to modernize and digitalize the sales operations of the electronic store. Currently, all sales activities are conducted manually, with limited customer support and no online purchasing capability. The OSS serves as a comprehensive replacement for the existing manual sales process by introducing an automated, centralized and user-friendly online platform.

The system will interact with both customers, recipients, store staff and web administrators through a web interface. It will also contain a database server that stores product information, user accounts, purchase records and order details. On the management side, it will also contain databases to store staff accounts, admin accounts, system related settings and audit information. In future releases, the system can be upgraded further to integrate with external payment gateways, courier APIs, and order tracking.

### 2.2 Product Features

The Online Selling System provides a range of core functionalities designed to improve both user experience and store efficiency. Major features include:

- User Account Management: Customers can register, log in, and manage personal profiles and purchase histories.
- Product Browsing and Filtering: Users can view products by category, brand, or technical specifications and apply filters for easy comparison.

- Product Details View: Each product can be displayed in higher details such as model, brand, price, release year, warranty period and specifications, etc.
- Shopping Cart and Checkout: Customers can add items to a virtual cart, proceed to checkout and select payment as well as delivery information.
- Recipient support: Order details will be sent via emails and phone number to the recipient for confirmation.
- Storage management: Store staff can  product listings, monitor orders, update inventory and conduct sales.
- Administrative Management: Admin can manage user roles, change system settings as well as conduct system updates and maintenance.

## 2.3 User Classes and Characteristics

The Online Selling System is designed to serve multiple user classes, each with distinct roles, access privileges, and levels of interaction with the system. The following are the anticipated user classes and their characteristics:

- Customers:
  - Description: End-users who browse and purchase electronic products through the online platform.
  - Characteristics:
    - May use the website at anytime of the day
    - Use the website occasionally or frequently based on shopping habits.
    - Customers will only access the customer side of the website, this is independent of the admin and staff side.
    - Access to general system functions such as account creation, product browsing, filtering, searching and purchasing.
    - Varying level of technical expertise, from beginner to expert.
    - Require an intuitive and user-friendly interface for easy navigation.
  - Special consideration: Accessibility and usability are the top priorities for this user class. As the end-users of the application, this class is the most favored of all.
- Recipients:
  - Description: End-users who are the recipients of the orders bought from the website.
  - Characteristics:

        - They do not need to interact with the website through an online platform.
        - They only need to be informed about the orders' information.
    - Special considerations: This type of end-users are often customers at the same time, so the characteristics of customers are also applicable here.
- Store staff:
    - Description: End-users who handle product listings, stock updates, and order processing within the system.
    - Characteristics:
        - Use the website at almost all times of the day due to working schedule.
        - Staff users have access to the staff side of the website, which handles the product related actions, including adding, updating or deleting product information, verifying stock levels, and processing customer orders.
        - Access limited to sales management and product maintenance functions.
        - Often have moderate technical skill levels.
        - Require an user-friendly interface for product management.
    - Special consideration: Need simple, comprehensive, efficient product management interfaces. They also need pre-training and technical manuals to help solve errors.
- Administrative:
    - Description: End-users who have the highest privilege in the system.
    - Characteristics:
        - Use the website at regular intervals, mostly in business hours.
        - Admins have unlimited access to everything in the websites, which also include system settings and other user's roles.
        - Their action needs to be audited for accountability.
        - Have high technical ability.
    - Special considerations: Require advanced control features, secure access, and comprehensive reporting tools.

## 2.4 Operating Environment

The Online Selling System is expected to operate in the following environments:

- Hardware: Any modern computer, smartphone and tablet capable of running web browsers.
- Operating system: Platform-independent (Windows, macOS, Linux, Android, iOS).
- Browser Requirements: Compatible with major browsers such as Google Chrome, Microsoft Edge, Firefox, and Safari.

## 2.5 Design and Implementation Constraints

- Technical constraints:
  - The system must be web-based and responsive across devices.
  - The database must ensure ACID compliance for transaction integrity.
  - The system must support HTTPS for secure communication.
  - All the users data in the database must be safely protected during both storing and retrieving.
  - All third-party libraries or frameworks must be stable, actively maintained, and properly licensed to avoid legal or dependency issues.
  - The backend must use reliable, open-source technologies (e.g., PHP/Laravel, Node.js, or Python/Django) that are compatible with the customer's hosting infrastructure.
- Operational constraints:
  - The system must operate efficiently for 100 concurrent user sessions.
  - The software must follow predefined coding standards and document practices as it will be maintained by the customer's organization.
  - System deployment will be performed on the customer's designated hosting server and configuration changes will follow the organization's IT policies.
- Design constraints:
  - Consistent interface design following usability and accessibility standards.
  - The architecture should be modular and loosely coupled, allowing individual modules (e.g., Product Management, Order Tracking) to be updated without affecting the entire system.
  - Must allow future integration with payment gateways and delivery service APIs.

## 2.6 User Documentation

The following user documentation, provided in digital format (PDF and web-based) and following IEEE 1063 – Software User Documentation, will be delivered along with the system:

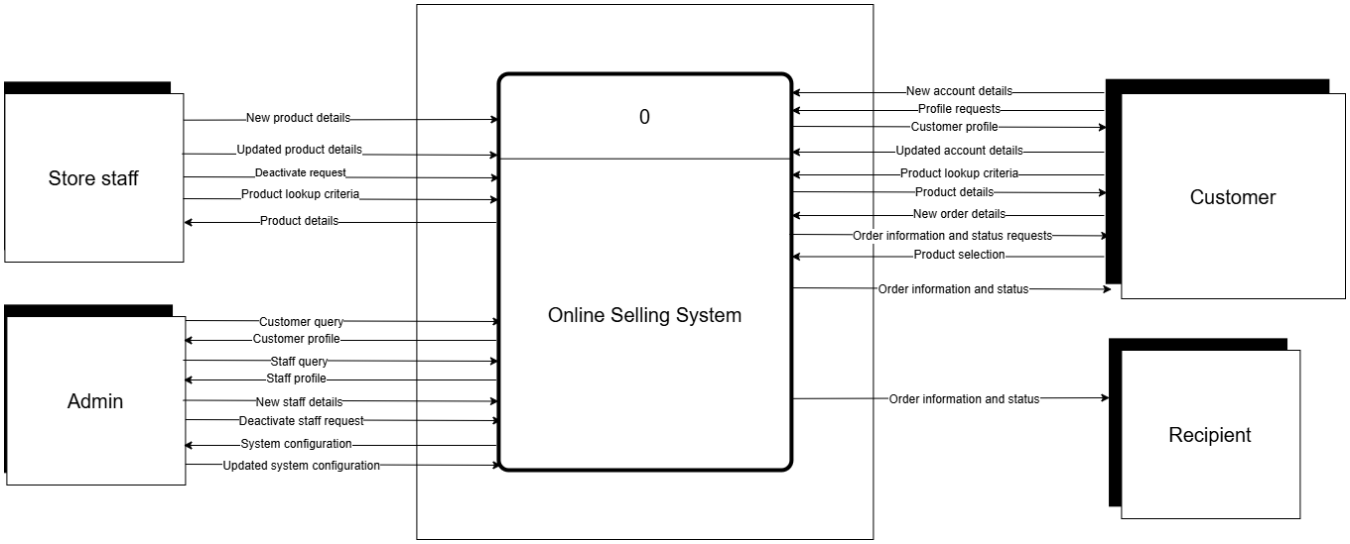
- User Manual: Detailed instructions for customers on how to register, browse and make purchases.
- Staff Guide: Step-by-step documentation for managing products and solving product management related errors.
- Administrative Guide: Step-by-step documentation for users management and system configurations.

## 2.7 Assumptions and Dependencies

- The system assumes users have stable internet access and basic familiarity with online shopping applications (such as Shoppee, Lazada).
- The server hosting environment will be continuously available and properly maintained.
- The product database must be inspected and cross-validated regularly by store administrators to ensure accurate information.
- The system depends on a web framework and database platform that are supported and updated by their vendors.
- Both the staff and the administrators are trained carefully before operating the system.

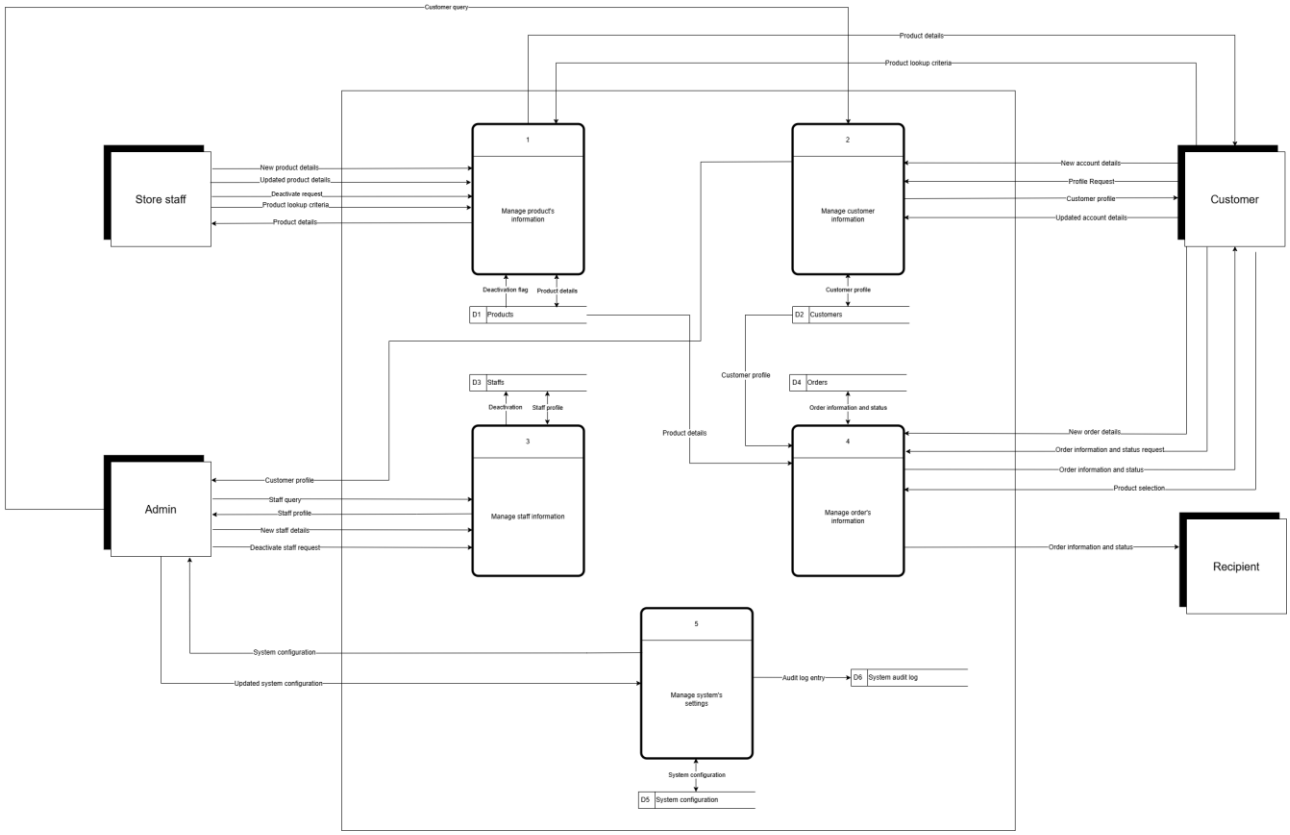
# 3. Data Flow Diagrams

## 3.1 Context Diagram




The context diagram illustrates the highest level of abstraction for the Online Selling System, representing the entire system as a single process (Process 0) and identifying all external entities that interact with it. Four external entities have been identified and modeled: Store Staff, Customer, Admin, and Recipient. This diagram establishes the system boundary and scope, documenting all data exchanges between external entities and the system without exposing internal architectural details or process decomposition.

## 3.2 Diagram 0




The Level 0 diagram provides the first-level functional decomposition of the Online Selling System, breaking down the single context-level process into five major processes, six data stores, and their interconnecting data flows.

Process 1 (Manage product's information) encompasses all product catalog management operations. Store Staff provides new product details and updated product details as inputs, and submits product lookup criteria to query the inventory. The process stores and retrieves product information from data store D1 (Products) and returns product details to both Store Staff and Customers upon request.

Process 2 (Manage customer information) handles customer account lifecycle management. It receives new account details from Customers during registration, processes customer profile requests bidirectionally, and manages updated account details for profile modifications. Customer records are persisted in data store D2 (Customers), which also provides customer profile data to Admin for administrative oversight.

Process 3 (Manage staff information) manages staff accounts and authorization. Admin submits staff profile information and updated staff role assignments to this process, which maintains staff records in data store D3 (Staffs).

The process also returns staff profile data and customer profile information to Admin as needed.

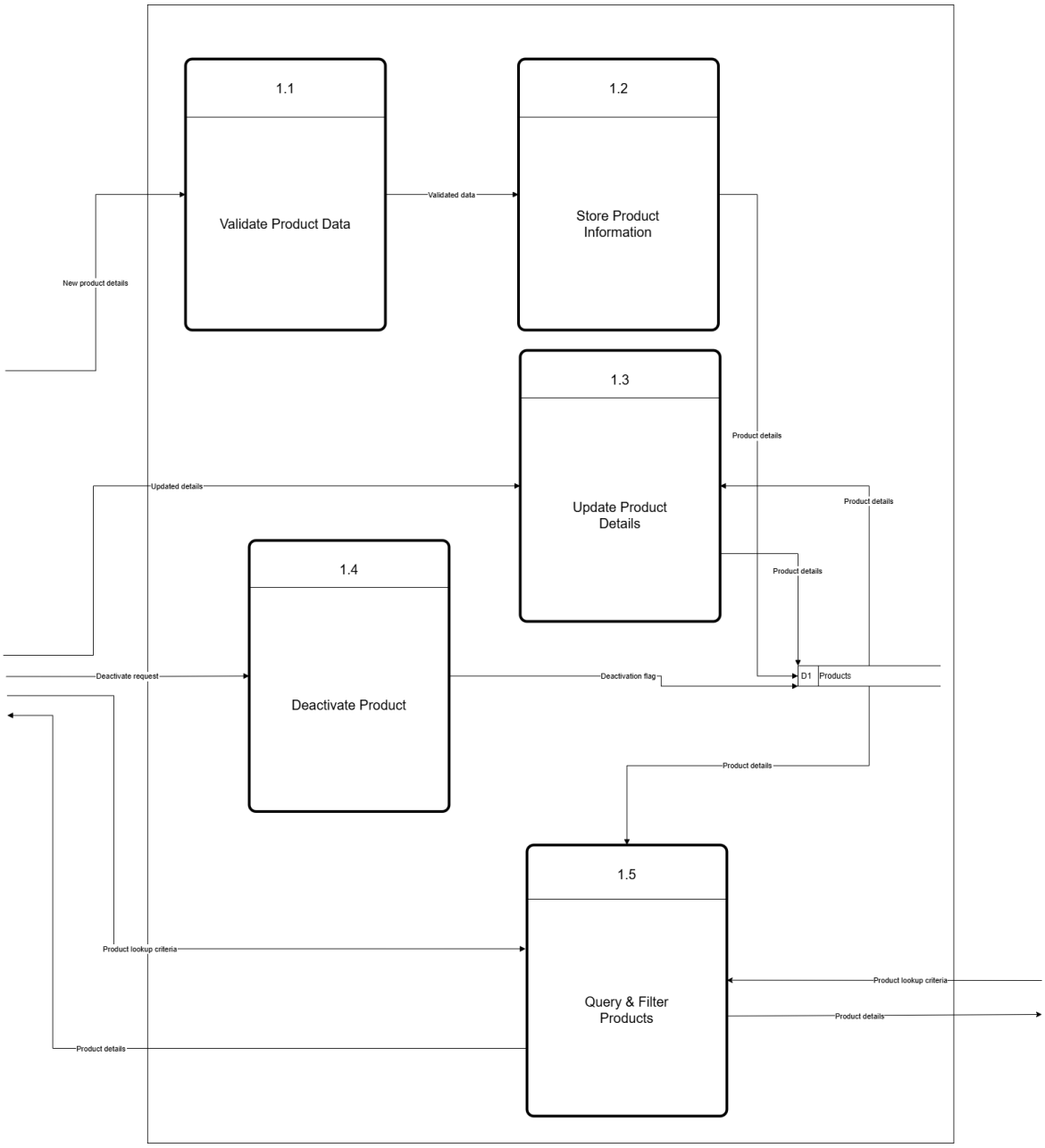
Process 4 (Manage order's information) orchestrates the complete order management workflow. It receives new order details from Customers, reads product details from D1 (Products) to validate cart contents and calculate pricing, accesses customer profile data from D2 (Customers) for order association, and stores completed orders in D4 (Orders). Upon order creation, the process transmits order information and status updates to both Customers and Recipients. This process implements the core transaction functionality of the system.

Process 5 (Manage system's settings) provides system configuration and audit capabilities. Admin sends system configuration parameters and updated system configuration to this process, which persists settings in data store D5 (System configuration). All configuration changes generate audit log entries that are written to data store D6 (System audit log), ensuring traceability and compliance with audit requirements.

The diagram demonstrates clear separation of concerns across functional domains while illustrating data dependencies through shared data stores. Notably, D1 (Products) serves as a reference for order processing, D2 (Customers) links customer accounts to orders, and D6 (System audit log) captures system-wide change events for administrative oversight.

## 3.3 Child Diagrams (and so on)

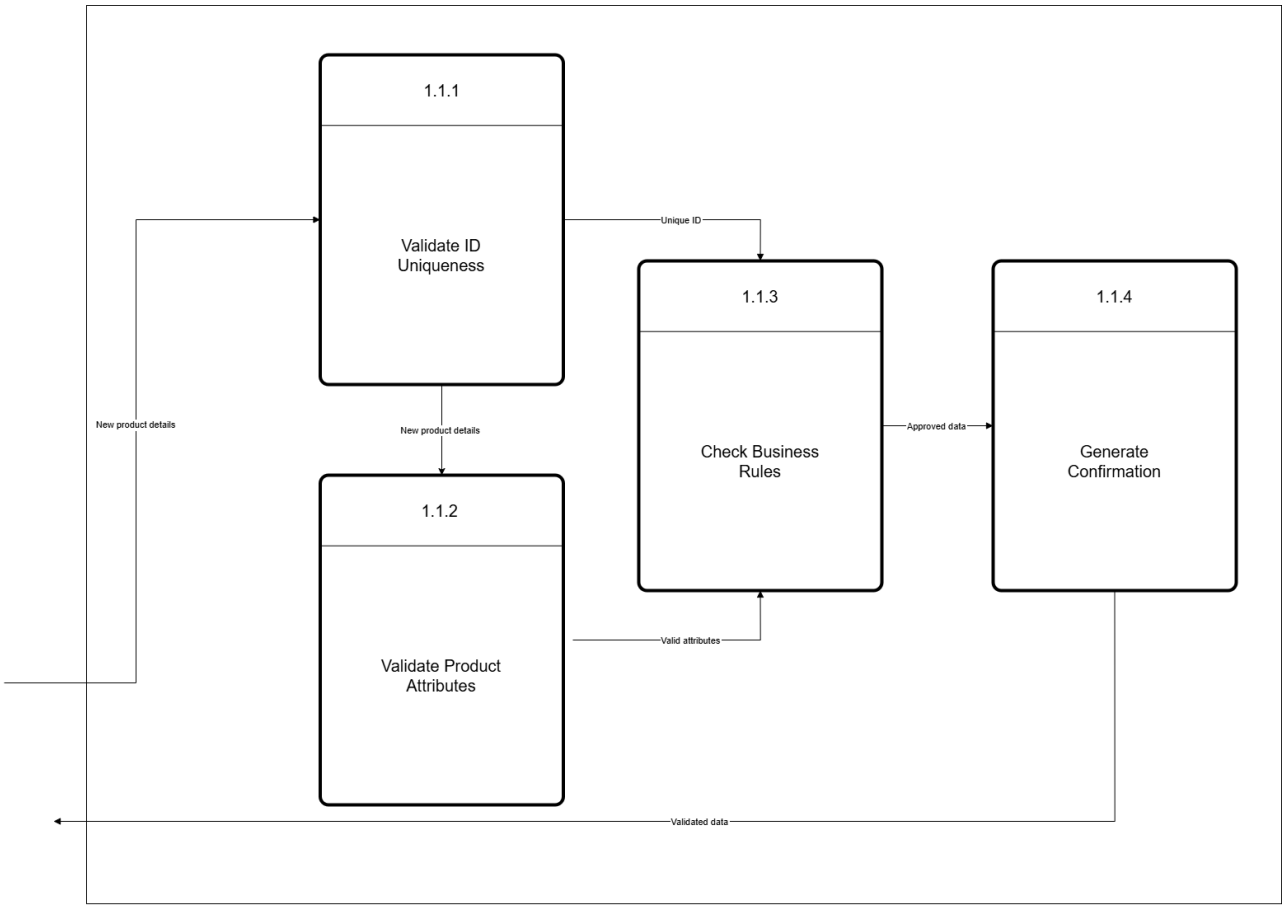
### 3.3.1 Manage product's information (Process 1 – Level 1)




This diagram represents the processes involved in managing product data within our system. It illustrates how store staff are able to add, update, deactivate, and search for products, while customers can retrieve product information for viewing. The main data store, D1 Products, holds all product records and acts as the central repository for this subsystem. Key processes include validating product data,

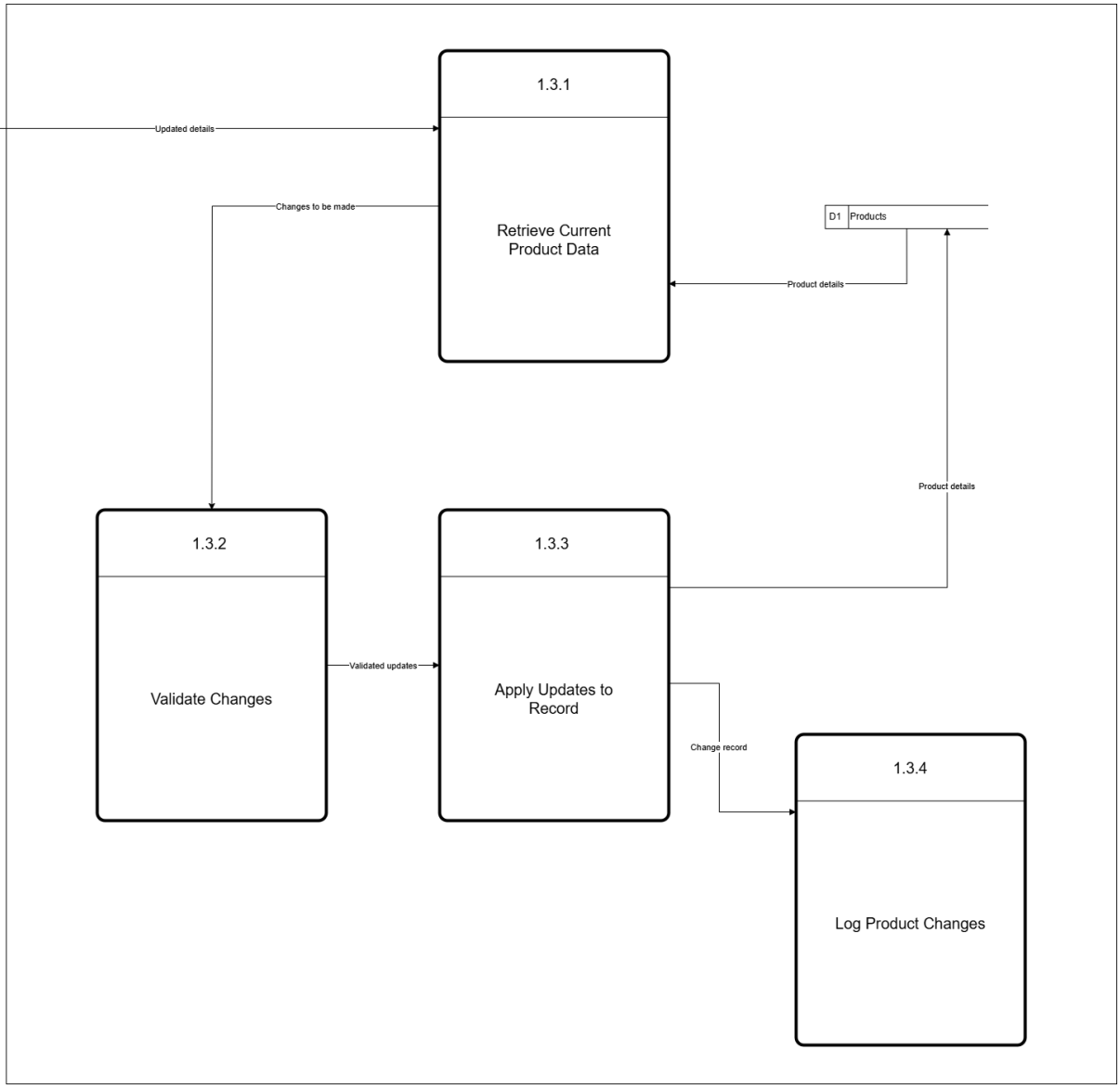
storing new records, updating product details, and querying the database. Each process ensures that all product information entering or leaving the system is properly verified to maintain data accuracy and integrity. This DFD demonstrates how the product management function supports both administrative operations and customer access within the overall system.

#### *3.3.1.1 Add New Product (Process 1 - Level 2)*



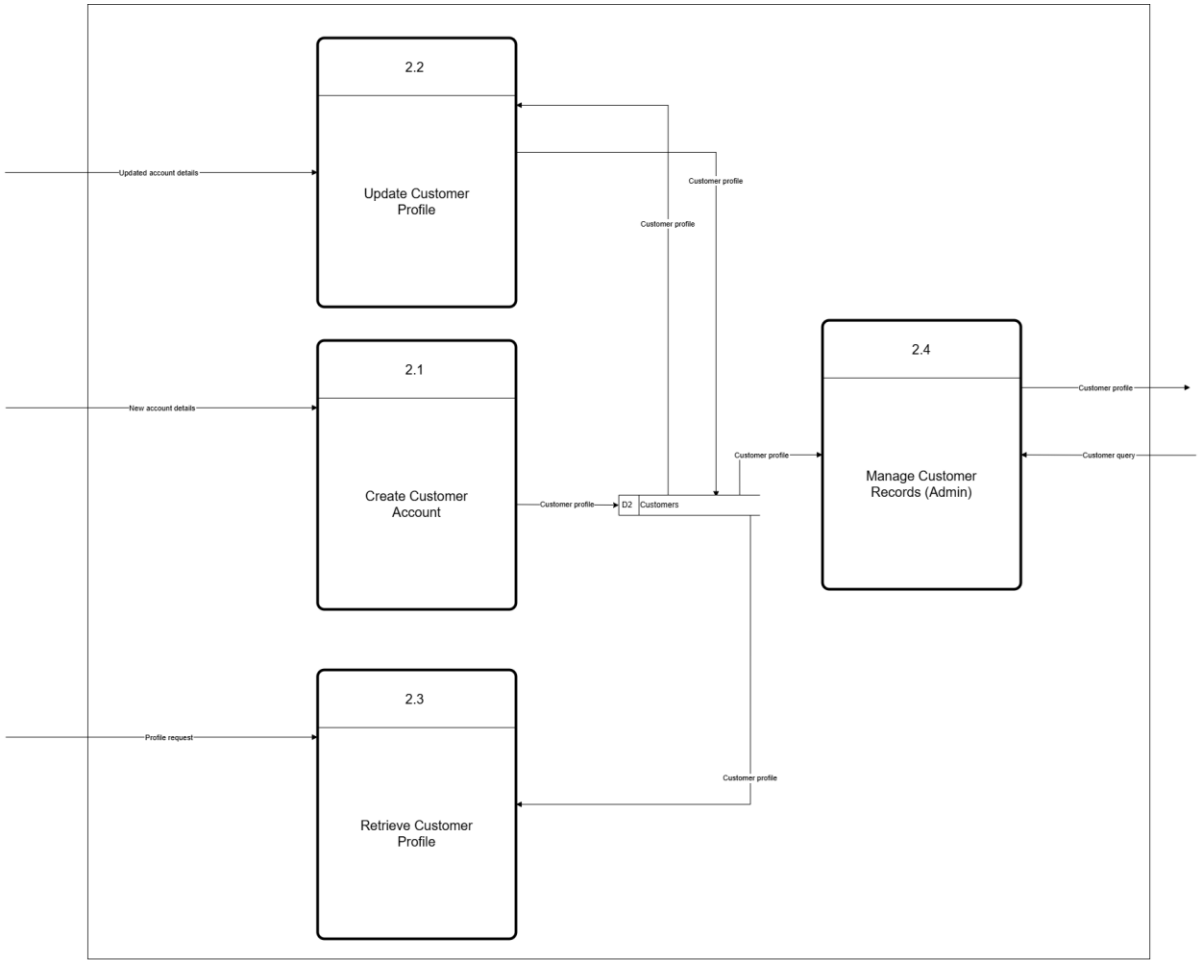

This diagram provides a detailed view of the process for adding new products to the system. It outlines the step-by-step validation flow beginning with checking product ID uniqueness, verifying product attributes, and enforcing business rules before final confirmation and storage. The validated data are then saved to D1 Products, ensuring that only accurate and approved information is recorded. This level of detail was included to demonstrate how the system safeguards against duplicate entries and invalid product data, reinforcing the reliability of the product catalog and maintaining consistency across the platform.

### *3.3.1.2 Update Product (Process 1 - Level 2)*



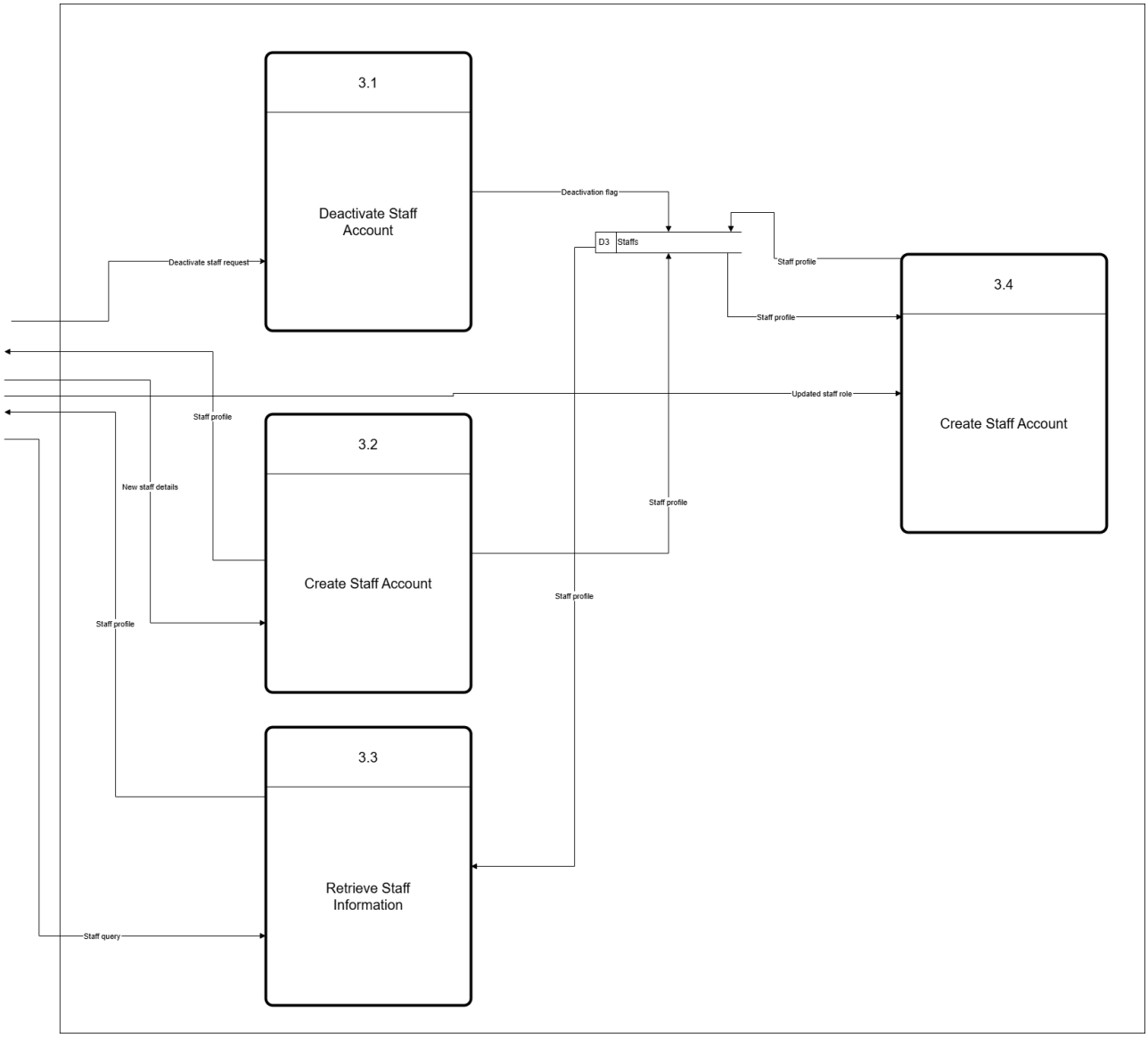

This Level 2 DFD expands on the update process for existing products. It illustrates how the system retrieves product information from D1 Products, validates any proposed changes and applies approved updates. This flow was designed to ensure that all product updates are performed in a controlled manner. The diagram highlights the verification mechanisms that support data accuracy and accountability, ensuring that the product database remains both current and audible.

### 3.3.2 Manage Customer Information (Process 2 - Level 1)



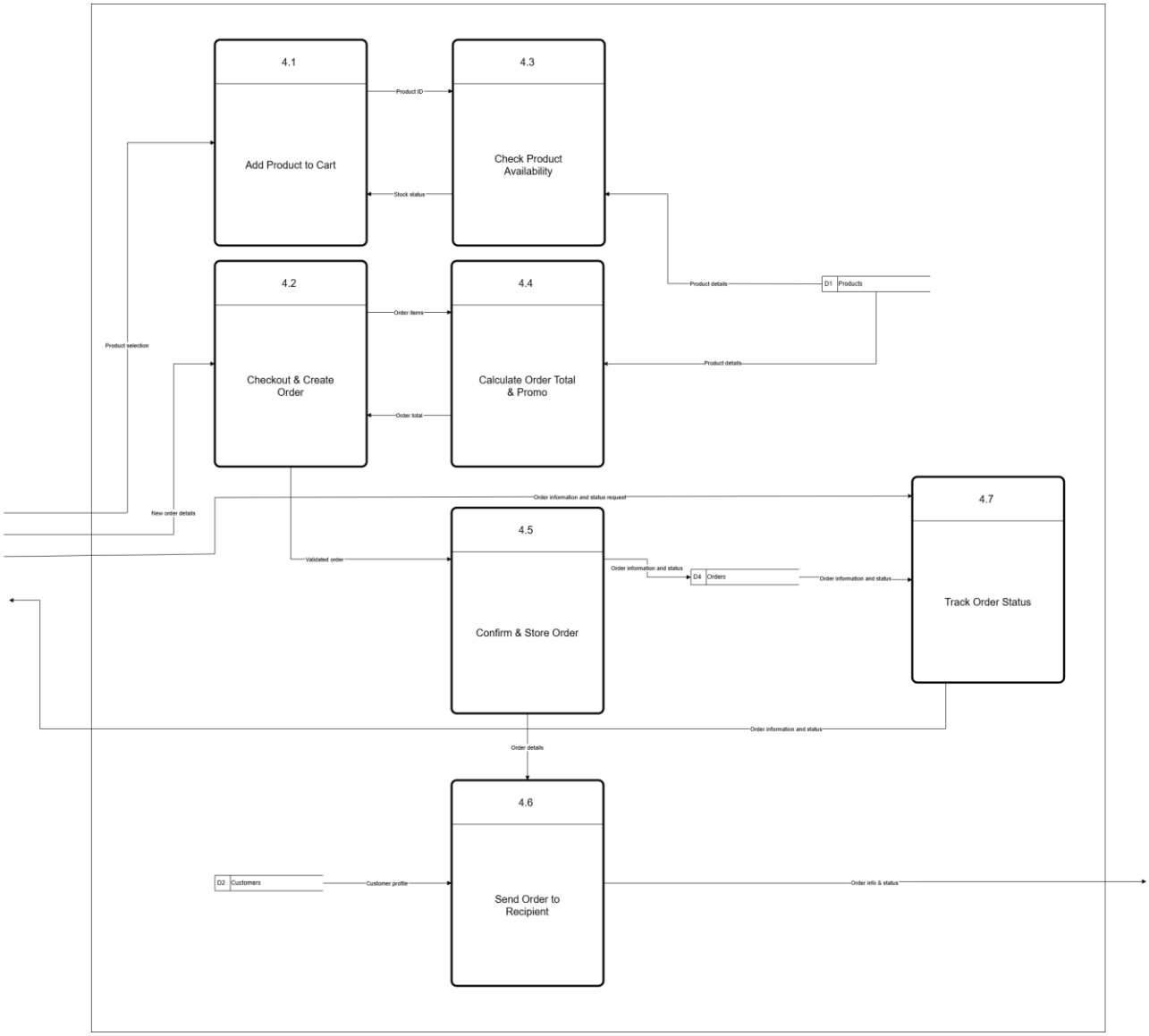

This diagram shows how the system manages customer-related data, including registration, profile updates, information retrieval, and authentication. The external entities are the Customer, who interacts directly with the system, and the Admin, who has privileges to manage customer records. All customer data are stored in D2 Customers, which serves as the primary repository for personal and account information. Through this diagram, we aim to emphasize the system's ability to maintain accurate and secure customer information throughout its lifecycle.

### 3.3.3 Manage Staff Information (Process 3 - Level 1)



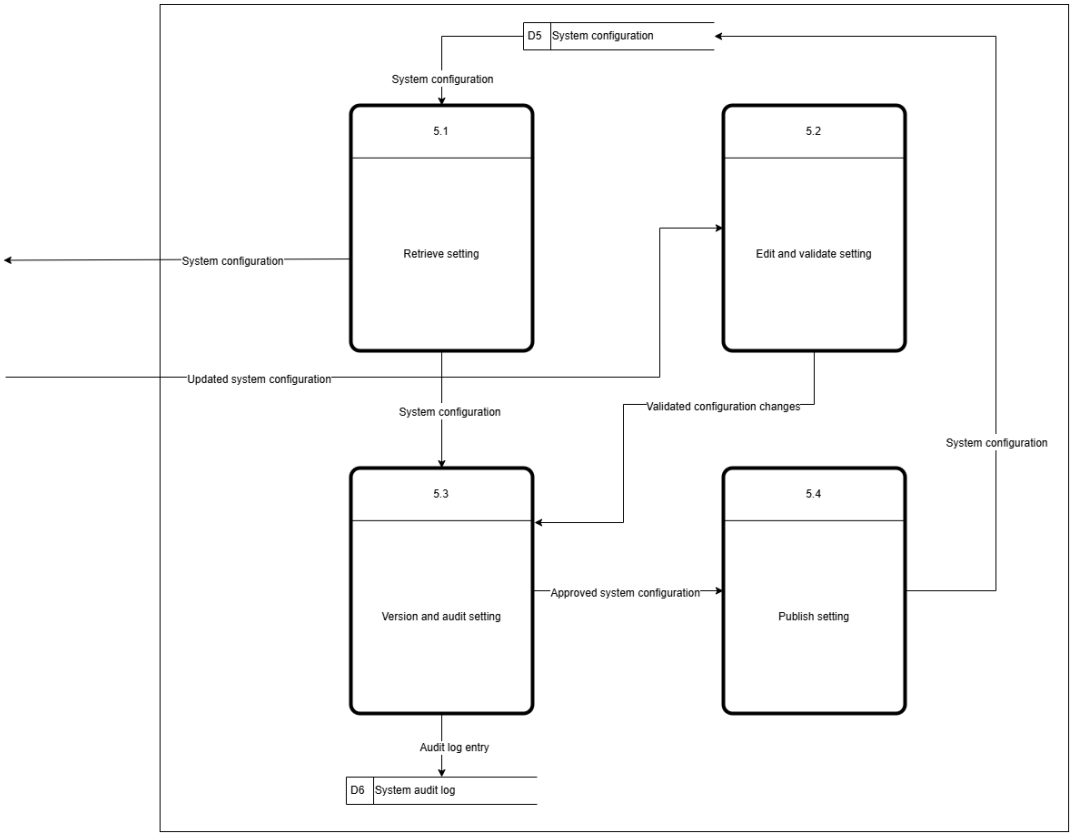

The D3 diagram outlines the management of staff accounts within the system. It includes processes such as account creation, role and profile updates, information retrieval, and account deactivation. The Admin entity oversees most staff management operations, while the Staff entity interacts mainly through login and authentication. All records are stored in D3 Staffs, the data store that maintains internal staff details. This diagram demonstrates how the system supports internal account administration and enforces role-based access control to maintain operational security and data consistency.

### 3.3.4 Manage Order Information (Process 4 - Level 1)




This diagram represents the complete flow of order management, from product selection to order tracking. The main external entities are Customer and Recipient, while the data stores involved include D1 Products, D2 Customers, and D4 Orders. The processes capture key actions such as adding products to the cart, checking availability, calculating totals and promotions, confirming orders, and sending order details to recipients. Each step ensures accurate synchronization between product inventory, customer details, and order records. This diagram demonstrates how data flows across different components of the system to facilitate a seamless and reliable ordering process.

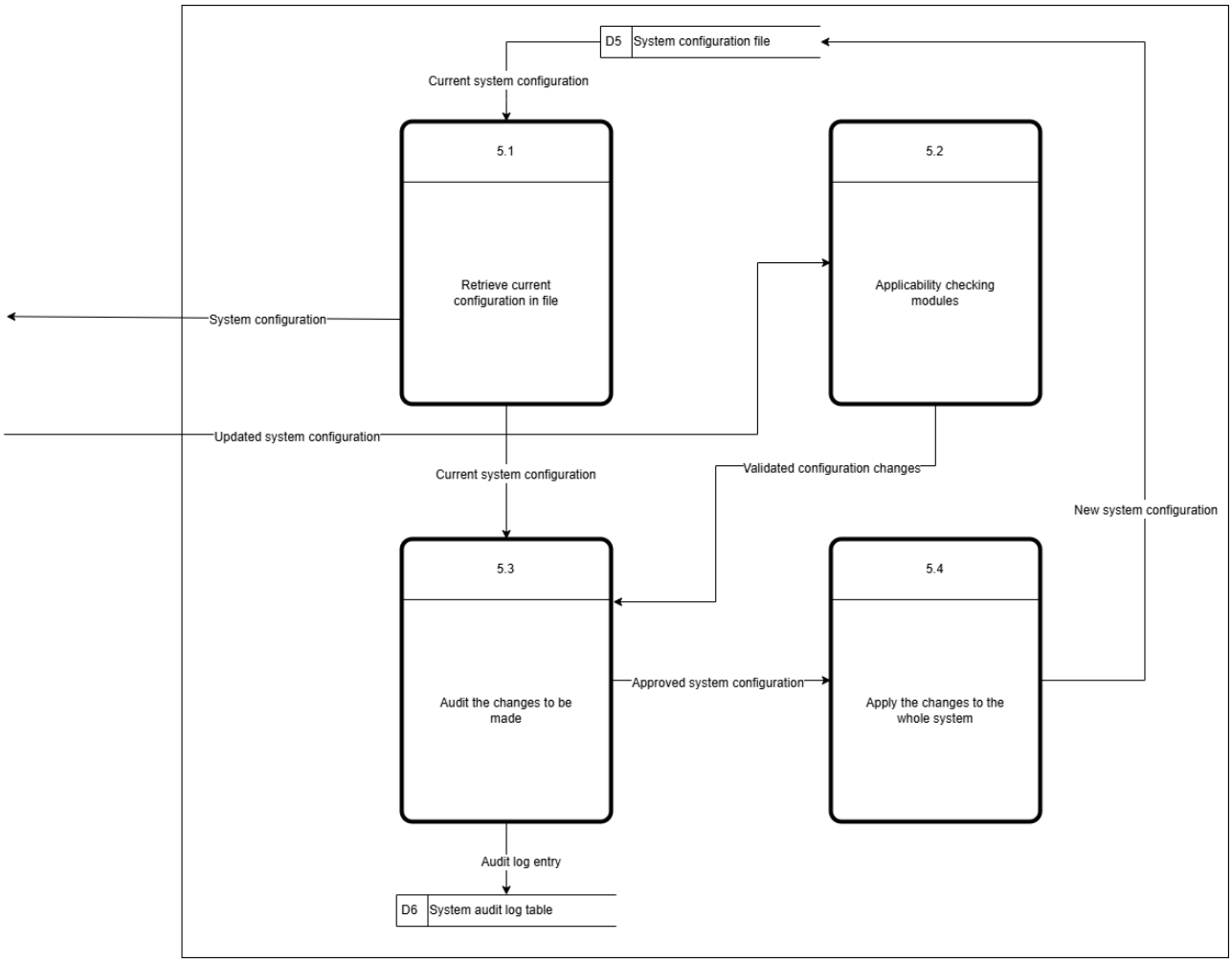
### 3.3.5 Manage System's setting (Process 5 - Level 1)




This diagram represents how the system handles configuration management. It shows the flow of retrieving, editing, validating, versioning, and publishing configuration settings. The main data store, D5 System Configuration, contains all system parameters, while D6 System Audit Log records any changes for tracking and accountability. The process begins with retrieving the current configuration, followed by editing and validation to ensure data accuracy. Once validated, the system versions and audits the changes before publishing the updated configuration. This diagram highlights how configuration updates are managed securely and systematically to maintain consistency across the system.

## 3.4 Physical Data Flow Diagrams (if needed)

Manage System's setting (Process 5 - Level 1)




This diagram illustrates the physical flow of managing system configuration files within the system. It outlines how configuration data are retrieved, validated, audited, and applied to the live environment. The main data store, D5 System Configuration File, contains the current and updated configuration data, while D6 System Audit Log Table records all modification activities. The process begins with 5.1 Retrieve Current Configuration in File, where the existing settings are accessed from the configuration store. The data are then passed to 5.2 Applicability Checking Modules, which validate and ensure that the configuration changes are compatible with the system. Once validated, 5.3 Audit the Changes to Be Made reviews and logs the changes for approval, ensuring transparency and traceability. Finally, 5.4 Apply the Changes to the Whole System implements the approved configuration updates.

This diagram demonstrates how configuration management is physically executed to ensure system reliability, proper validation, and audit compliance.

# 4. System Features

## 4.1 Core functions:

- The system shall store manufacturer-provided product IDs and ensure they remain unique.
- The system shall link each product ID with its attributes (brand, model, specifications, warranty, release year, price).
- The system shall ensure that when multiple customers attempt to purchase the last product, the order is given to the first confirmed checkout.
- The system shall create and keep track of immutable order records containing product details, price, promotions, and customer information at the time of purchase.

## 4.2 For all users (including guest users):

- The system shall allow users to browse products and view product details.
- The system shall allow users to filter and search products by attributes such as brand, model, specifications, warranty period, release year, and price.
- The system shall allow guest users to create a customer account.

## 4.3 For customers:

- The system shall allow customers to add products to a shopping cart.
- The system shall allow customers to view their order confirmation after checkout.
- The system shall allow customers to view their order status after placing an order.

### 4.4 For recipients:

- The system shall send the order information to the recipients after the checkout process via email or phone number.
- The system shall support cash on delivery (COD) as the payment method.

### 4.5 For admin:

- The system shall allow administrators to define and manage user roles (customer, staff, admin).
- The system shall allow administrators to manage system settings, update, and maintenance.

### 4.6 For staff:

- The system shall allow staff to add new products to the catalog.
- The system shall allow staff to update product details including attributes, price, and availability.
- The system shall allow staff to remove (deactivate) products from the catalog.

## 5. Nonfunctional Requirements

### 5.1 Performance Requirements

- The system shall support at least 100 concurrent users without performance degradation.
- The system shall ensure an average response time of no more than 3 seconds for product browsing and checkout operations.

### 5.2 Safety Requirements

- The system shall operate continuously (24/7) with a minimum uptime of 95%.
- The system shall be able to recover from crashes within a developer-defined timeframe (RTO ≤ 30 minutes).

- The system shall allow regular maintenance activities with minimal downtime (under 2 hours).

### 5.3 Security Requirements

- The system shall comply with Viet Nam's data protection laws and regulations to secure customers' information.

### 5.4 Software Quality Attributes

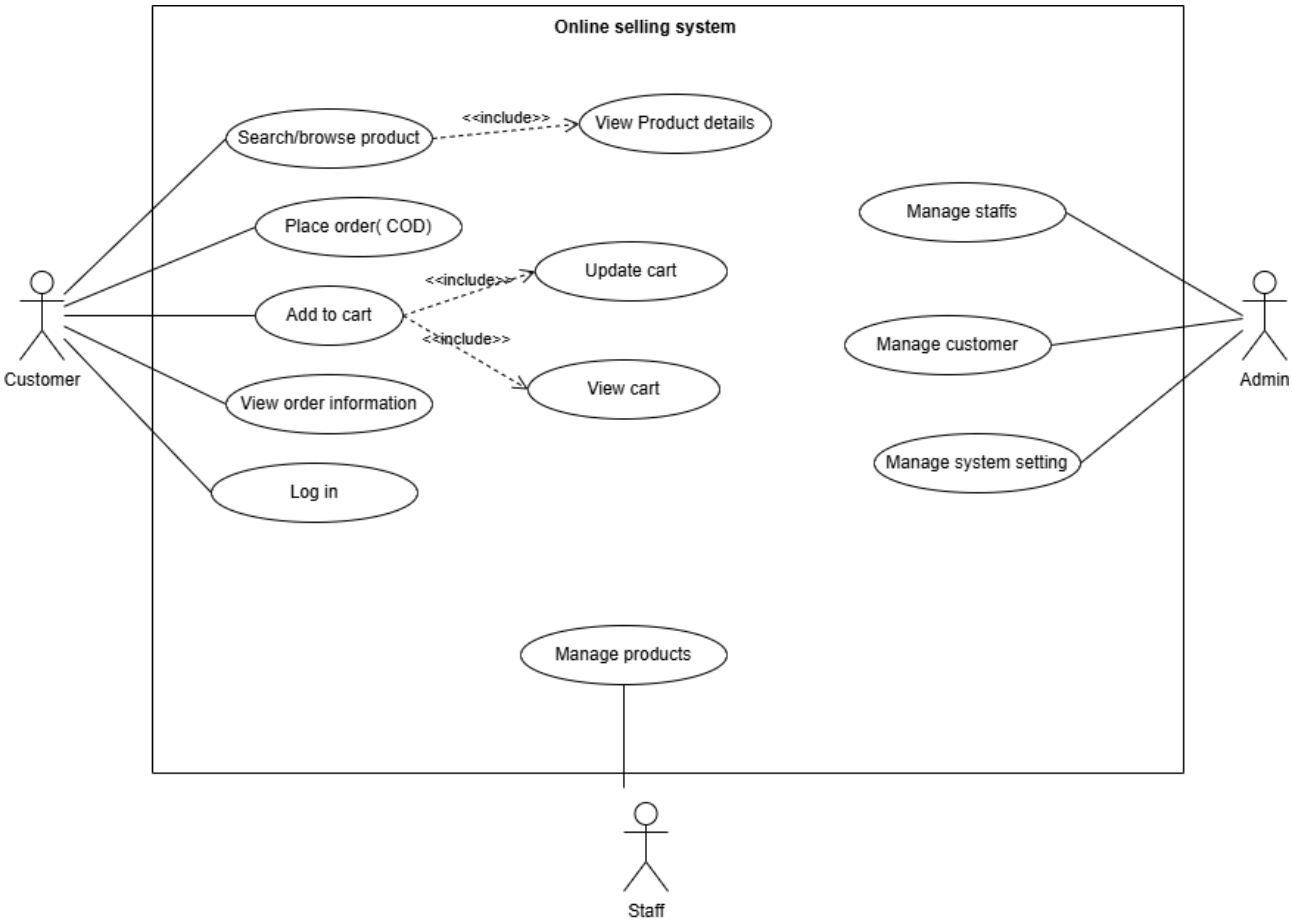
- The system shall provide a responsive user interface accessible on both desktop and mobile devices.
- The system shall provide clear navigation, error messages, and accessible forms for checkout and product browsing.
- The system shall be designed with modular architecture to support future extensions such as third-party delivery, inventory tracker integrations.
- The system shall allow the addition of new payment methods and delivery options in future releases without redesigning the core architecture.

## 6. Other Requirements (if needed)

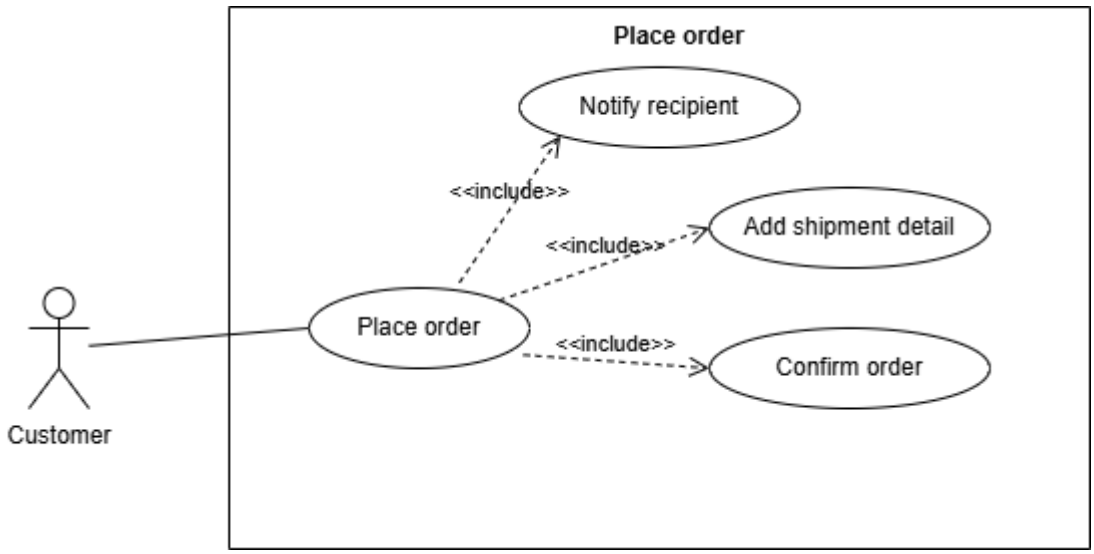
*None*

# 7. System Analysis

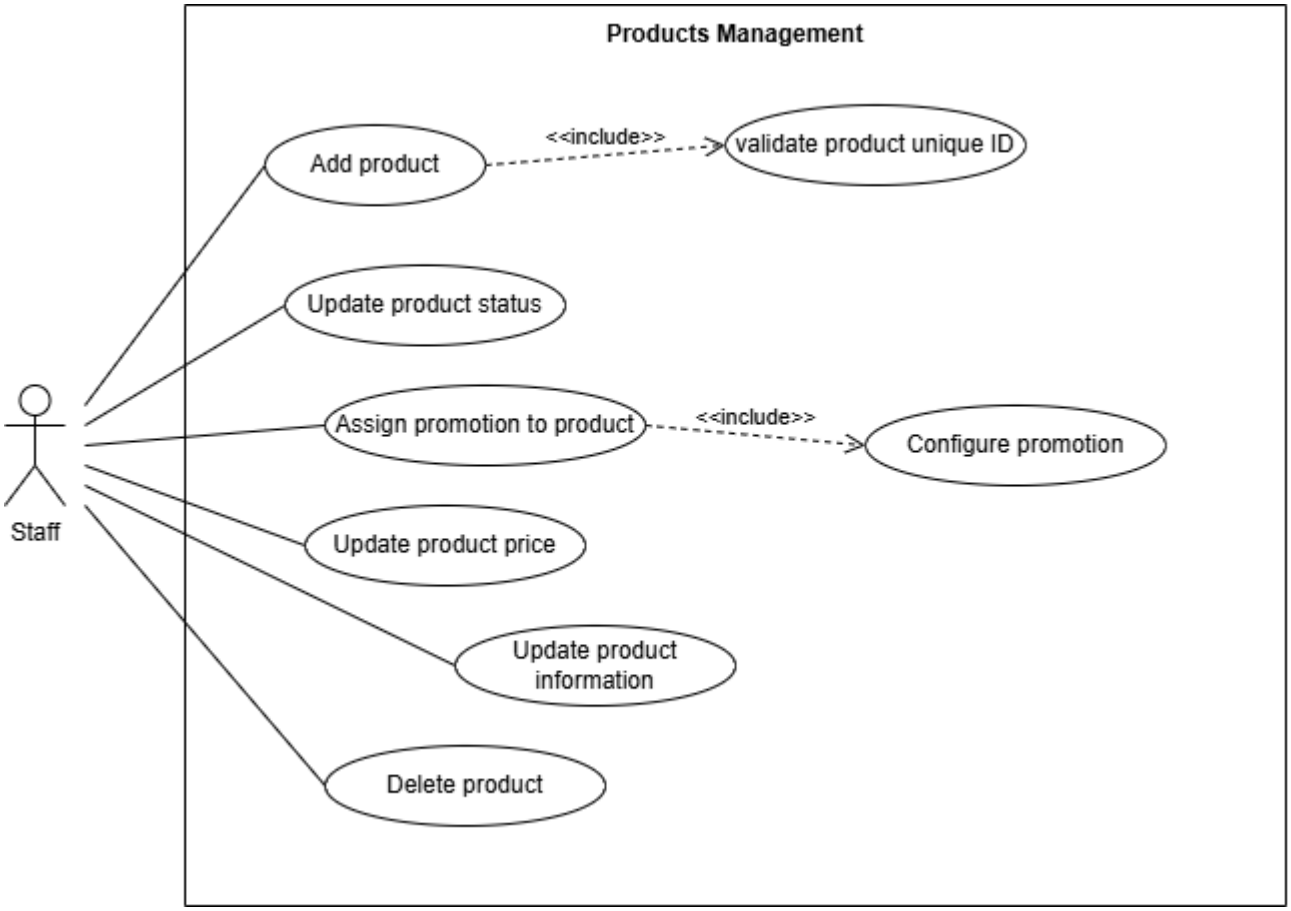
## 7.1 Use case Diagrams




This is a the use case diagram for the whole system. This use case diagram shows the interaction between main actors of the system. The application boundary encloses all behavior offered to three external actors. Customer represents the storefront user who explores the catalog, views product pages, manages the shopping cart,… Staff is the back-office operator responsible for Manage products and Manage orders, ensuring the catalog stays accurate and orders progress through the lifecycle. Admin governs platform-level data and options through Manage staffs, Manage customer, and Manage system setting.




The use case Place order includes two mandatory sub-steps: Add shipment detail collects the recipient's name, phone, address, and any delivery notes, and Confirm order captures the customer's explicit intent to proceed.



The Products Management use case includes capabilities a store staff member needs to curate the catalog. A staff user can create a new product, update core attributes (name, description, media), change business status (active/inactive), adjust prices, assign promotions and delete products.

## 7.2 Use case Scenarios

| | |
|---|---|
| Use case name | Place order |
| Description | Customer completes checkout by entering shipment details and confirming the purchase |
| Trigger | Customer clicks Place order on the checkout page |
| Primary actor | Customer |
| Pre-condition | Cart contains at least one purchasable item; checkout page is accessible. |
| Post-condition | New order is stored with a unique ID . |
| Normal flow | 1) System displays checkout.<br>2) Customer enters shipment details (recipient name, phone, address, note).<br>3) The system validates the details.<br>4) Customer confirms the order.<br>5) The system creates the order, assigns the order ID, persists the cart snapshot, records timestamps.<br>5) The system send order information to the recipient via email/SMS.<br>7) The system shows confirmation. |
| Alternative flow | A1) Shipment details invalid → system highlights field errors → customer edits and resubmits (returns to step 3).<br>A2) Customer abandons checkout before confirmation → no order is created; user may return to cart or storefront. |
| Exception flow | E1) Persistence failure during creation → system logs error, shows failure message, cart remains intact.<br>E2) Duplicate confirm click → system detects duplicate and returns |

| | |
|---|---|
| | the existing confirmation for the already created order. |

Table 1: Use case - Place order

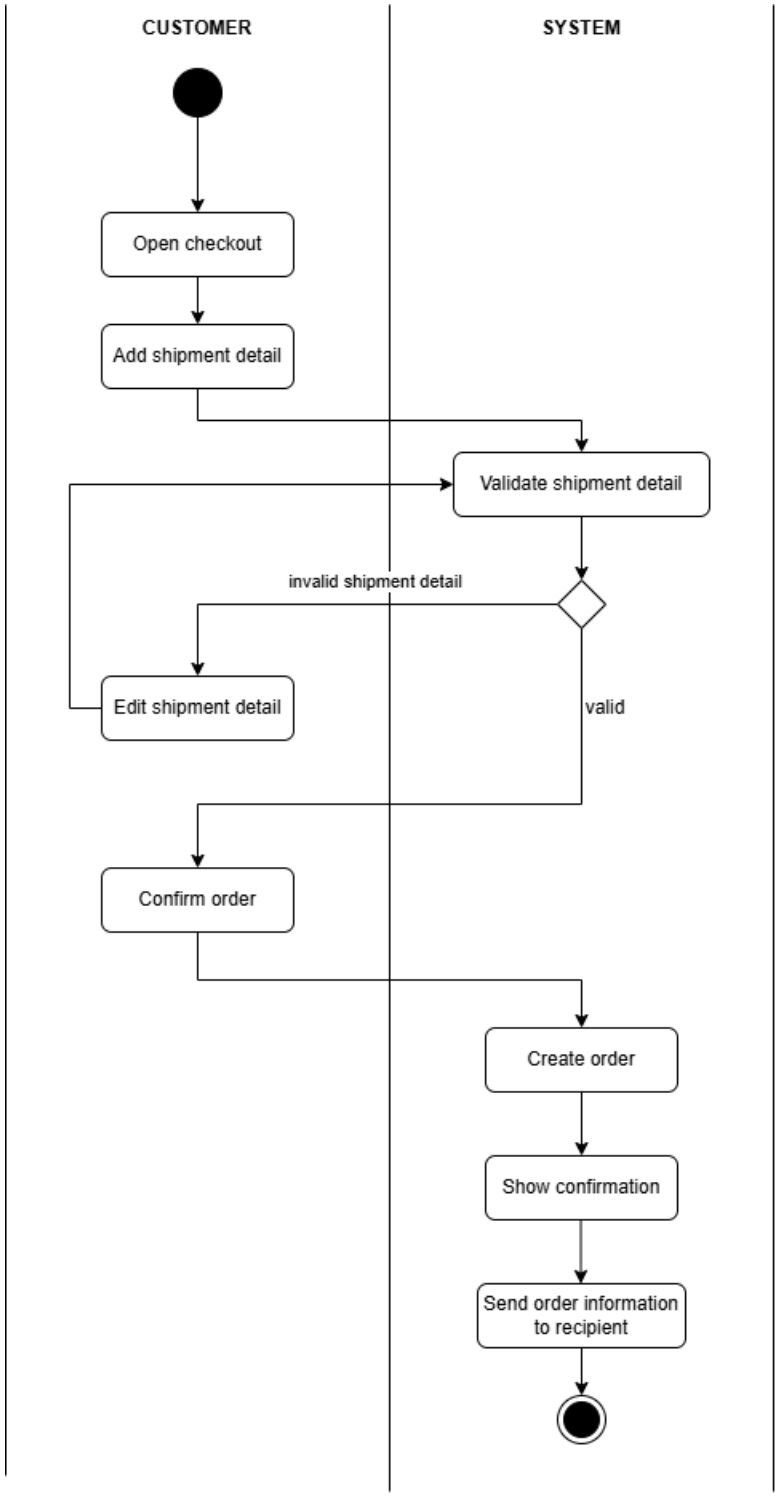
| | |
|---|---|
| Use case name | Add product |
| Description | A store staff member creates a new catalog item by entering required details and saving it to the system. |
| Trigger | Staff selects Add new product from the Products Management area. |
| Primary actor | Store Staff |
| Pre-condition | Staff is authenticated and authorized;The product creation form is available; required reference data (e.g., categories) exists. |
| Post-condition | A new product record exists with a unique ID |
| Normal flow | 1) UI displays the product creation form.<br>2) Staff enters product data and submits.<br>3) Controller validates required fields and formats.<br>4) Controller checks the uniqueness of ID.<br>5) On success, controller saves the record and returns the new product ID.<br>6) UI shows a success confirmation and may redirect to the list or detail page. |
| Alternative flow | A1 – Validation errors: invalid or missing fields → controller returns errors → UI highlights issues → staff corrects and resubmits (back to step 2).<br>A2 – Duplicate ID/SKU: uniqueness check fails → controller returns “duplicate” → UI prompts for a different ID/SKU → resubmit (back to step 2). |

| | |
|---|---|
| | A3 – User cancels: staff aborts the operation → no record is created; UI returns to the previous page. |
| Exception flow | E1 – Database failure: insert/transaction error → system logs the failure, rolls back changes, and shows a generic error; no product is created.<br>E2 – Service outage: validation or persistence services unavailable → system reports temporary error and suggests retry. |

Table 2: Use case- Add product

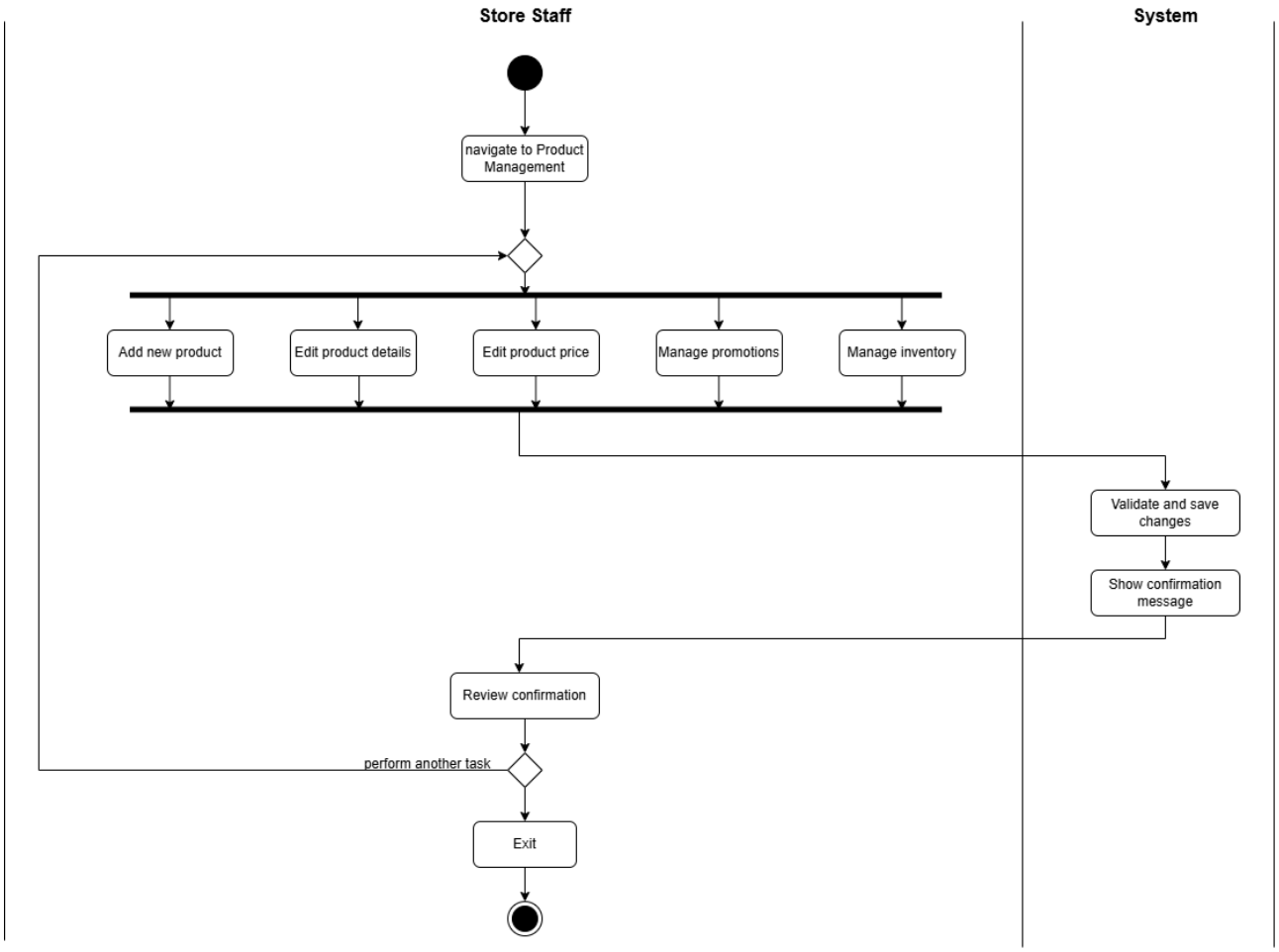
## 7.3 Activity Diagrams

### 7.3.1 Place order activity diagram



The customer opens the checkout, supplies shipment details, and awaits validation. The system verifies required fields and formats, returning field-level messages when something is missing or malformed, which sends the customer into a short correction loop until the data satisfy validation. Once the information is valid, the customer confirms the purchase; the system creates a new order in an initial state such as Placed or Pending, stores a snapshot of cart contents for traceability, and presents a confirmation containing the order identifier. After the order is created, the system sends an order information to the recipient via email/SMS.

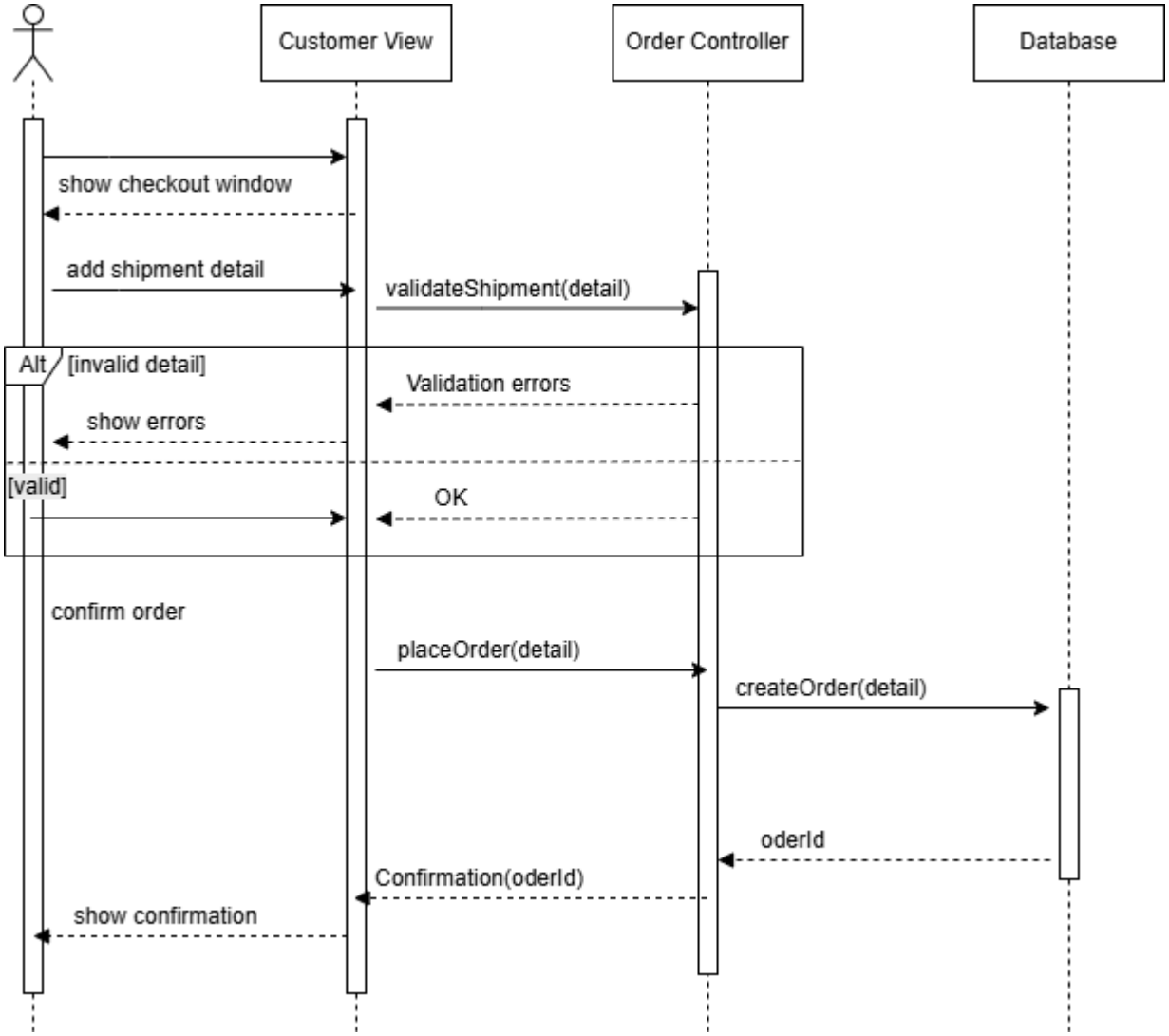
### 7.3.2 Product management activity diagram




The flow begins when a staff user opens the Product Management area. A decision point lets the user choose a single task—add a new product, edit product details, edit price, manage promotions, edit product status or delete a product. After the user completes the selected form, the system validates input, persists changes, and then shows a success confirmation. The user reviews the confirmation and either loops back to perform another task or exits.

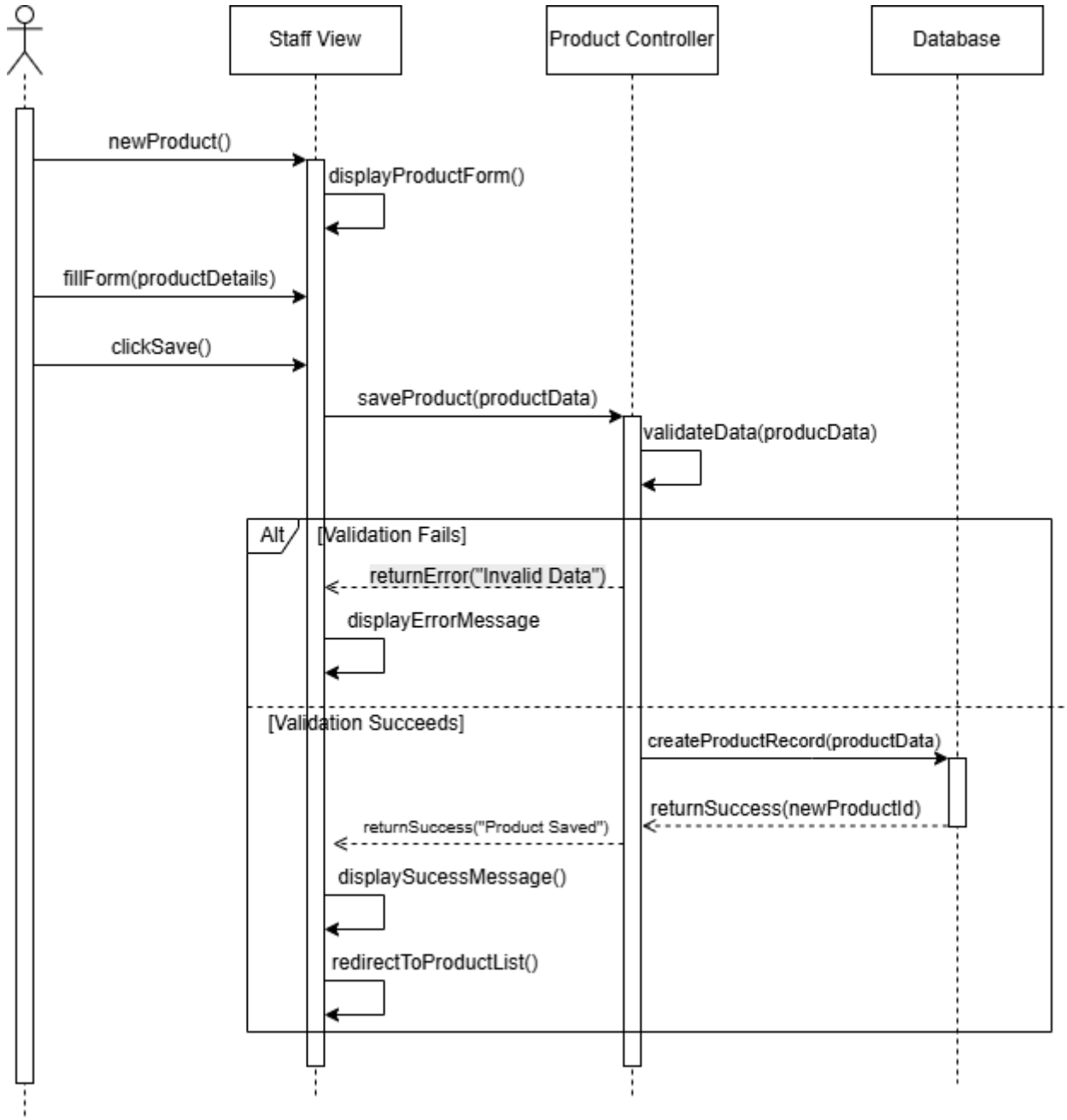
## 7.4 Sequence Diagrams

### 7.4.1 Place order sequence diagram




The view collects shipment information and calls the controller to validate it. An alternative fragment clarifies outcomes: invalid input triggers a validation-error response that the view renders back to the customer, whereas valid input leads the view to request order creation. The controller commits the order through the database, obtains the new identifier, and returns a confirmation that the view displays.

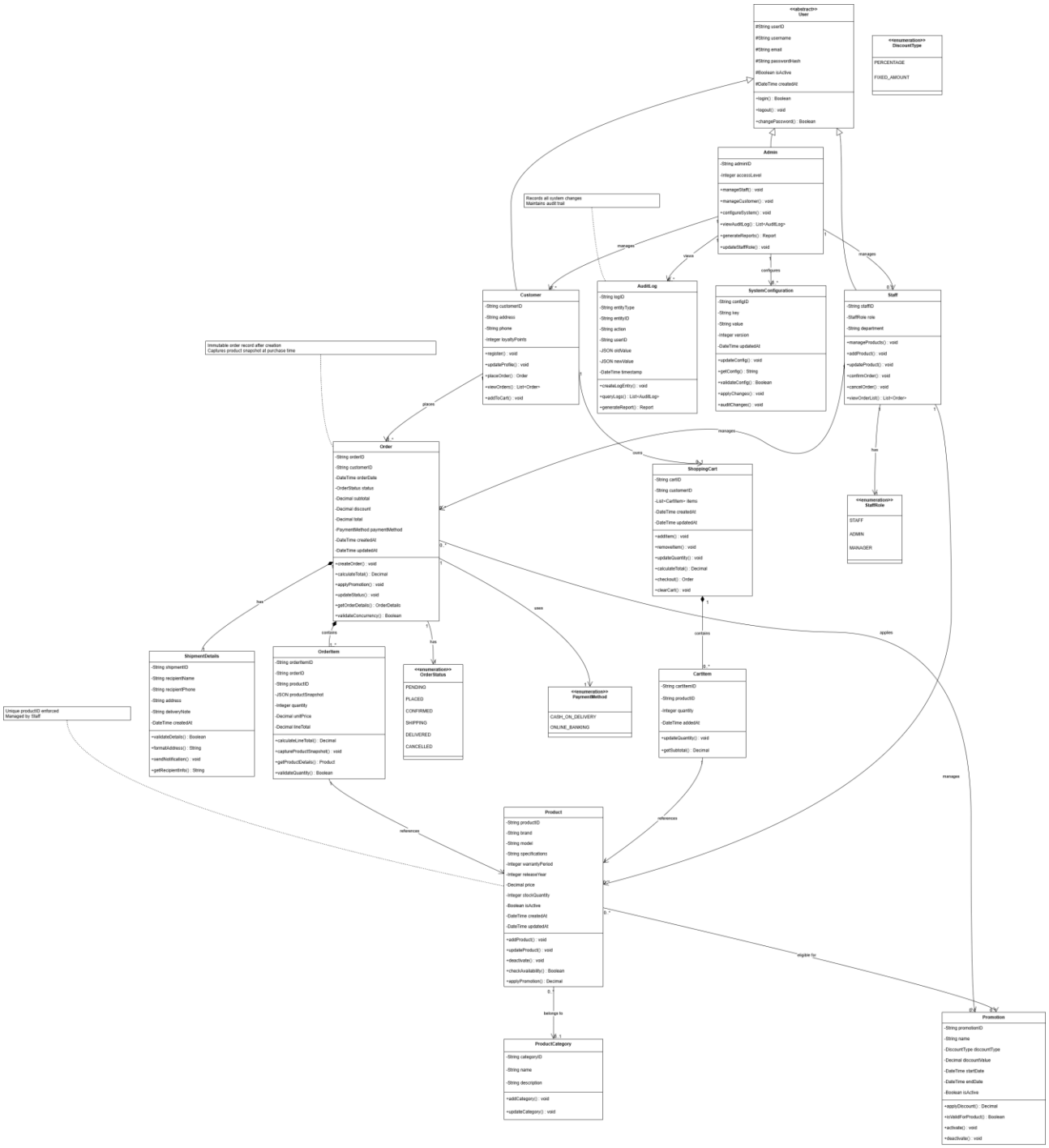
### 7.4.2 Add product sequence diagram




A staff user opens the Add product function. The controller validates all fields and performs a mandatory uniqueness check against the database for the product ID. An alternative fragment captures two paths: when validation fails, the controller returns field-level errors and the UI highlights them for correction; when validation succeeds, the controller inserts the new product, receives the generated identifier,

and returns a success response. The UI then shows a confirmation message and typically navigates back to the product list or into the new product's detail page.

## 7.5 Class Diagrams




The class diagram presents a layered architecture organized around core business entities and their interactions. A user hierarchy establishes role-based access

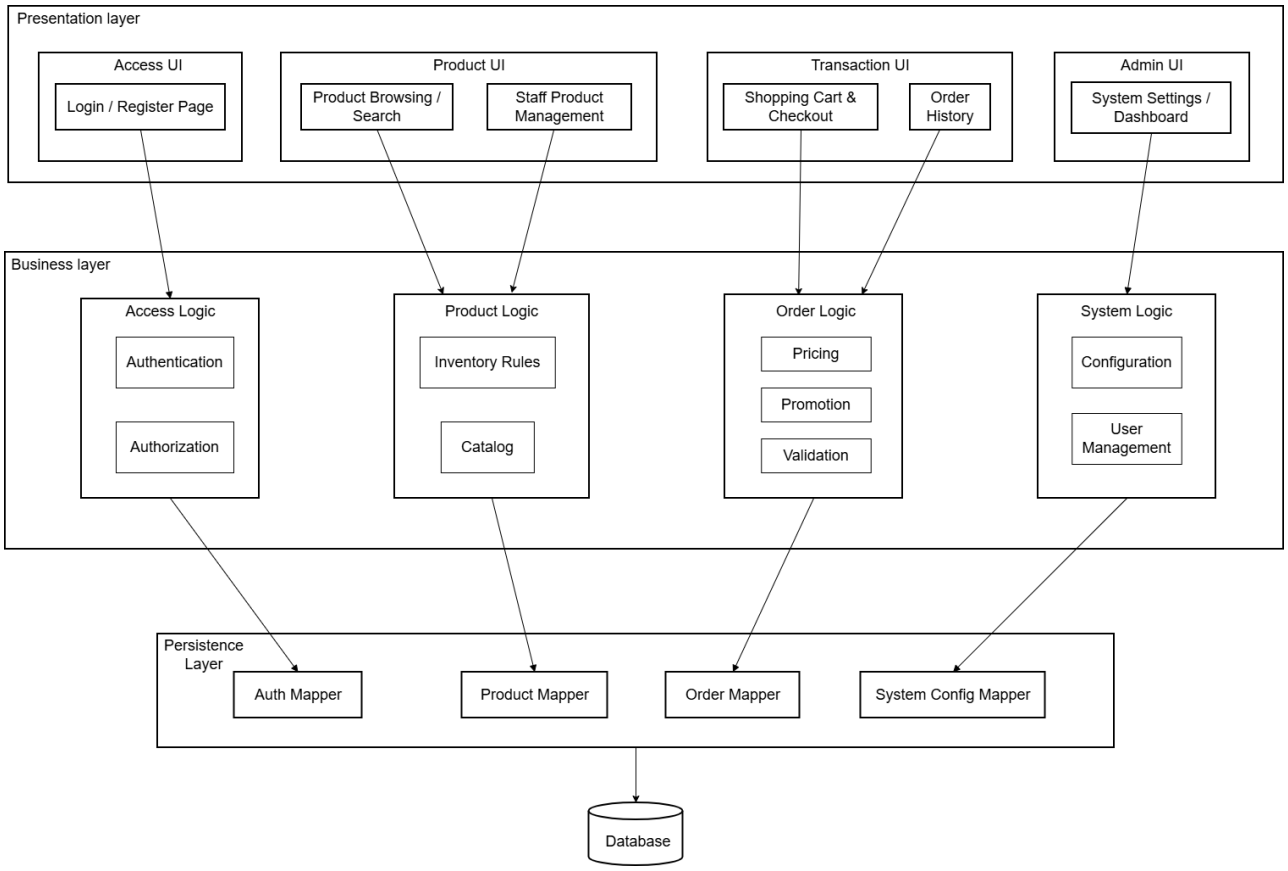
through inheritance, enabling different actors to perform distinct system operations. The product domain encapsulates catalog items with their attributes and management capabilities. The order domain implements a compositional structure where orders aggregate line items and delivery information, maintaining data immutability for historical accuracy. Shopping functionality allows users to accumulate items before finalizing purchases. Supporting classes provide system-level services including change tracking and configuration management. Relationships between classes reflect standard e-commerce patterns with appropriate cardinalities, ensuring data integrity and supporting concurrent operations. The design adheres to object-oriented principles, separating concerns across domains while maintaining clear interfaces for future extensibility and integration with external systems.

## 7.6 Other Diagrams (if needed)

*None*

# 8. System Design

## 8.1 Architectural Design




The Online Selling System uses a three-layer architectural model, consisting of the Presentation Layer, the Business Logic Layer, and the Persistence Layer, all supported by a central relational database.

**Presentation Layer:** The top layer acts as the interface between the user and the system. It is responsible for capturing user interactions and displaying information. This layer is segmented into functional modules:

- **Access UI:** Manages user entry points such as the Login and Registration pages.
- **Product UI:** Handles the display of the product catalog for customers and provides management interfaces for staff.

- **Transaction UI:** Facilitates the shopping experience, including the Shopping Cart, Checkout process, and viewing Order History.
- **Admin UI:** Provides a dashboard for system administrators to manage settings and view system status.

**Business Logic Layer:** All functional rules and processing workflows are implemented in this layer. It is divided into four logical modules that correspond to major system domains:

- **Access Logic:** authentication and authorization handling.
- **Product Logic:** catalog operations and inventory rule enforcement.
- **Order Logic:** pricing, promotions, and validation of customer orders.
- **System Logic:** configuration management and user administration.

**Persistence Layer:** This layer provides data access components (Mappers) dedicated to each logic domain. These mappers are responsible for retrieving, updating, and storing data in the database while isolating business rules from low-level data operations.

## 8.2 Database Design

The Enhanced Entity–Relationship Diagram (EERD) organizes the system's data into clear, well-defined entities that capture all essential information for user management, product handling, and transaction processing.

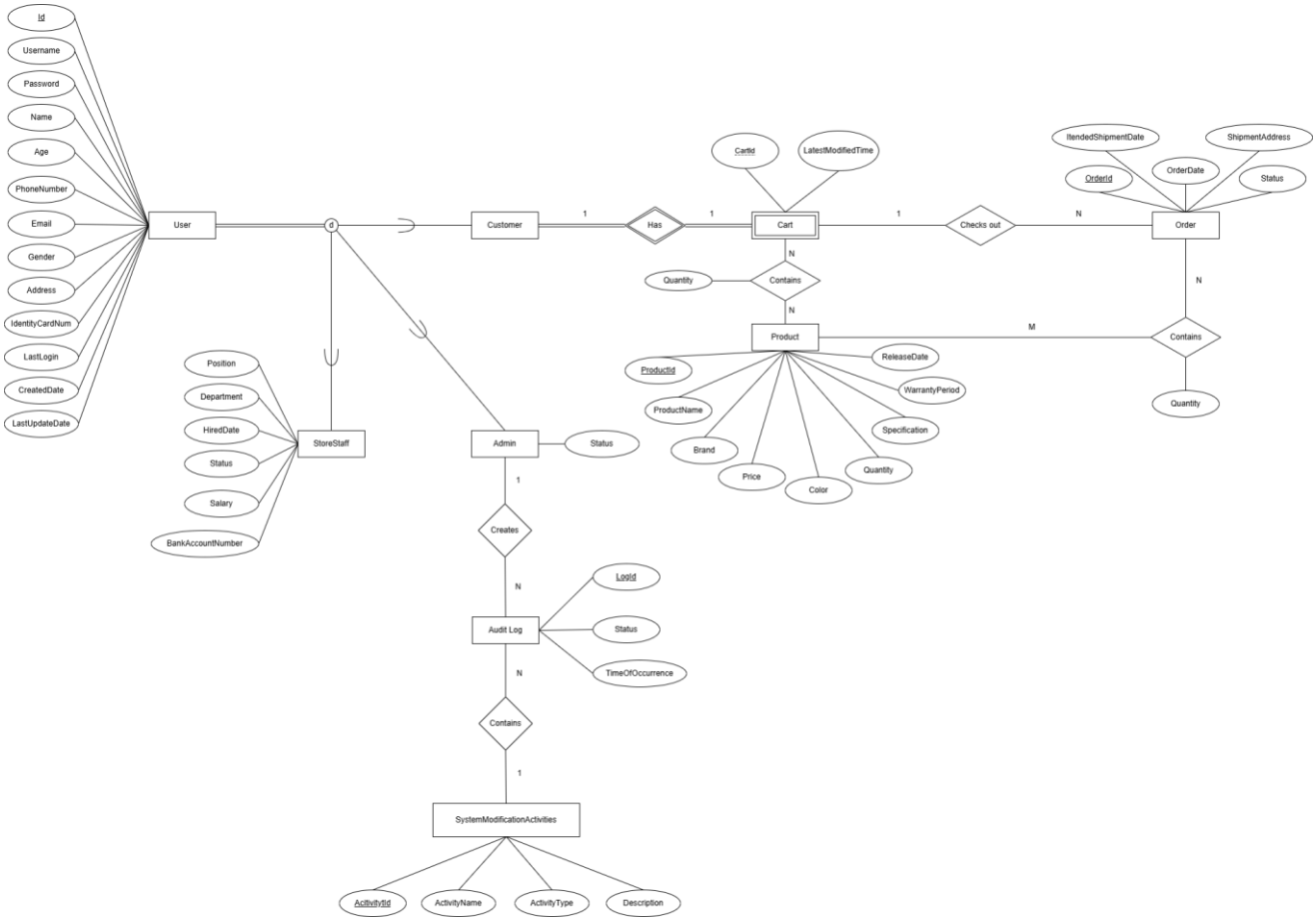
The model defines separate entities for different type of system users: Admin, StoreStaff, Customer, each carrying additional role-specific data such as position, salary, department, and status.

Core business data are represented through Product, Cart and Order. The Product entity stores detailed product information including brand, price, color, specifications, and warranty. Customer activity is captured through a Cart, which stores cart metadata and connects to products via a many-to-many "Contains" relationship with quantity tracking. Completed purchases are stored in the Order entity, which records order dates, shipment details, and ordered products through another many-to-many relationship. The information about recipients are also captured in the shipment details attribute of the Order.

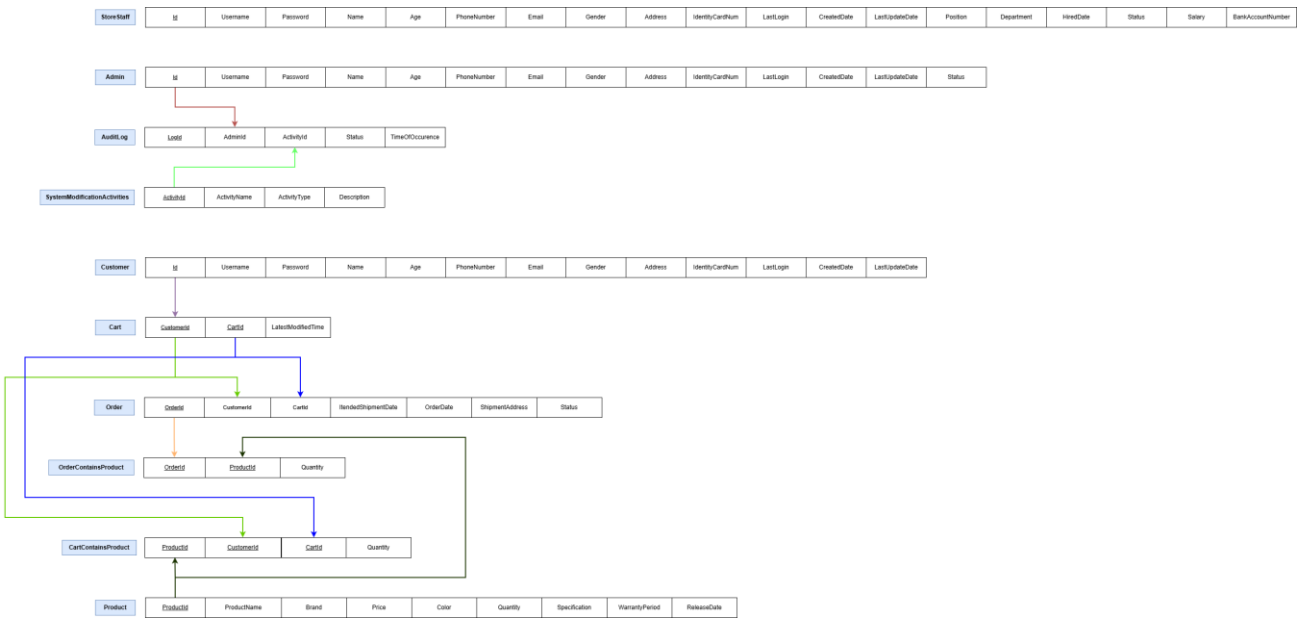
Besides core business data, system governance data is also represented in the system. They are showed as AuditLog and SystemModificationActivities, allowing admins to record operational actions and track detailed system changes.

Overall, the EERD provides a comprehensive structure ensuring all user data, product details, transactional records, and system activities are consistently captured and linked.

The EERD is as follow:




From the Enhanced Entity-Relationship Diagram above, we can create the database schema for the relational database of the company online electronic selling system as followed:




## 8.3 UI Design

To provide a clearer visualization of the proposed system interface, the complete Figma UI design can be accessed via the following link:

System UI design

# 9. Conclusions

In summary, this document sets clear, testable requirements for OSS v1.0 and aligns it with the store's business goal: moving from manual sales to a reliable, user-friendly online operation. It translates that goal into end-to-end customer journeys and role-based back-office workflows, anchored by a modular architecture that can grow to payments, delivery, and analytics without reworking the whole system. Quality targets include performance, availability, security, and usability are defined upfront so design, implementation, and testing can converge on measurable outcomes rather than assumptions. Assumptions, constraints, and documentation needs are also made explicitly to reduce delivery risk and streamline onboarding.

With these foundations, development can proceed with traceable requirements, QA can validate against objective criteria, and stakeholders can judge product success by improved customer experience, reduced manual workload, and expanded reach. This document is the contract for v1.0—and the roadmap's first step for future enhancements.